



Course Code and Name: INSY 8311 Database Development with PL/SQL

Installing Oracle Database XE and SQL Developer

Before You Start

- ✓ You need two separate downloads: Oracle Database XE (the database server) and SQL Developer (the client tool that connects to it). SQL Developer alone cannot run queries without a database to connect to.
- ✓ Install Oracle Database XE FIRST, then SQL Developer. Installing them in the other order is why you may not see a password prompt or be able to run sqlplus.
- ✓ You need about 3 GB of free disk space total and a stable internet connection.
- ✓ Oracle Database XE requires administrator rights to install. SQL Developer does not.

Part A: Install Oracle Database XE (the database server)

1. Go to the Oracle Database XE download page

Open a browser and navigate to: oracle.com/database/technologies/xe-downloads.html — this is a different page than the SQL Developer download page.

2. Download the installer for your OS

Choose "Oracle Database 21c Express Edition (21c XE) for Microsoft Windows x64" for Windows, or the equivalent Linux RPM for Linux. The file is around 2.5 GB — much larger than SQL Developer.

3. Accept the license agreement

Check the box to accept the license agreement before the download link becomes active.

4. Extract and run the installer

Unlike SQL Developer, this ZIP contains an actual installer. Extract it, then double-click setup.exe (Windows) to launch the installation wizard.

5. Approve administrator access

Windows will show a UAC prompt ("Do you want to allow this app to make changes?") — click Yes. This install does require admin rights.

6. Follow the wizard and set your SYS/SYSTEM password

Accept the license agreement, keep the default install location, and when prompted, set and confirm a password. This becomes the password for both the SYS and SYSTEM accounts. Write it down immediately — you'll need it constantly.

7. Wait for installation to finish

This takes 10–20 minutes as it sets up the database files, configures the listener, and creates the XE service. Don't close the window even if it looks stuck.

8. Finish and verify

Click Finish on the completion screen. The installer also adds sqlplus to your Windows PATH automatically. Close any open Command Prompt windows, open a new one, and type `sqlplus /nolog` — it should now be recognized. If it still isn't, log off and back on (or restart) so the PATH change takes effect.

Note: If you already installed Oracle Database XE previously and remember your SYS/SYSTEM password, skip Part A and go straight to Part B.

Installing Oracle SQL Developer

Before You Start

- ✓ You need about 500–600 MB of free disk space and a stable internet connection.
- ✓ Administrator rights are NOT required — SQL Developer runs from an unzipped folder, no installer needed.
- ✓ Have your database connection details ready (hostname, port, service name/SID, username, password) if you'll connect to a shared classroom database.

Step 1: Download SQL Developer

1. Go to the official download page

Open a browser and navigate to: oracle.com/tools/downloads/sqldev-downloads.html (or from the Oracle Developer Downloads page: Developer Tools → SQL Developer)

2. Pick the download for your operating system

The current release is SQL Developer 26.2. Choose the file that matches your computer:

- ✓ Windows (recommended): sqldeveloper-26.2.0.186.2220-x64.zip — includes JDK 17, nothing extra to install
- ✓ Windows (advanced): sqldeveloper-26.2.0.186.2220-no-jre — smaller download, but you must already have JDK 17 installed
- ✓ Mac (Apple Silicon – M1/M2/M3/M4): sqldeveloper-26.2.0.186.2220-macos-aarch64.app — includes JDK 17
- ✓ Linux: sqldeveloper-26.2.0-186.2220.noarch.rpm — does NOT include a JDK; install JDK 17+ first

3. Accept the license agreement

Check the box to accept the license agreement — the download link will not work until you do. Then click the download link. The file is large (500+ MB), so it may take several minutes.

Step 2: Extract the Files

4. Locate the downloaded file

It's usually in your Downloads folder.

5. Windows/Mac: extract it to a permanent location

Windows: right-click the ZIP → Extract All. Mac: double-click the .app archive. Extract to a location you won't accidentally delete — for example C:\sqldeveloper on Windows, or your Applications/home folder on Mac. Do not run SQL Developer directly from the Downloads folder.

Note: There is no installer program for Windows/Mac ZIP downloads. SQL Developer runs directly from the extracted folder. Linux uses the RPM installer — see the Linux steps below.

Step 3: Launch SQL Developer

6. Windows/Mac: open the extracted folder

Find the sqldeveloper folder or .app you just extracted.

7. Windows/Mac: start the application

Windows: double-click sqldeveloper.exe inside the folder. Mac: open the SQL Developer .app.

8. Linux: install and run from the terminal

Run these commands in order:

- ✓ `sudo rpm -Uhv sqldeveloper-26.2.0-186.2220.noarch.rpm` (installs the package)
- ✓ `cd sqldeveloper` (go to the installed folder)
- ✓ `./sqldeveloper.sh` (launch SQL Developer)

9. Wait for first launch

The first startup can take 30–60 seconds while it initializes — this is normal.

10. Handle the JDK prompt, if it appears

If you downloaded a version with JDK included, you normally won't be asked for a Java path. If prompted (common on the "no-jre" Windows version or on Linux), enter the path to your installed JDK 17 — for example `C:\Program Files\Java\jdk-17` on Windows, or `/usr/java/jdk-17` on Linux.

Step 4: Create a Database Connection

11. Open the New Connection dialog

Click the green "+" icon in the Connections panel, or go to Connections → New Connection.

12. Fill in your connection details

Enter a connection name (anything you'll recognize), your database username and password, and the hostname, port (default 1521), and SID or service name provided by your instructor.

13. Test and connect

Click "Test" to confirm the details work, then click "Connect" (or "Save" then double-click the new connection).

Step 5: Verify Everything Works

14. Check the Connections panel

You should see your new connection listed with your schema objects (tables, views, etc.) underneath it once expanded.

15. Run a test query

Click the SQL worksheet icon (pencil and paper) and run: `SELECT 1 FROM DUAL;` If it returns a result, you're fully set up.

Troubleshooting Tips

- ✓ Connection fails / times out: double-check hostname, port, and that you're on the correct network (e.g., campus Wi-Fi or VPN if required).
- ✓ "Unable to find JDK": use the version with JDK included for your OS, or double-check your JDK 17 installation path.
- ✓ Windows app won't open / crashes on startup: this can be a missing or 32-bit `MSVCR100.dll`. Most computers already have the correct 64-bit version, but if not, install the Microsoft Visual C++ Redistributable or update your `PATH` so a 64-bit copy is found first.
- ✓ Mac says the app is from an unidentified developer: right-click the app → Open, then confirm in the dialog that appears.
- ✓ Linux: `rpm` command fails: make sure you're running it with `sudo` and that JDK 17+ is installed before running `sqldeveloper.sh`.

SQL Developer Configurations

Part 0: System User Login

Understanding CDB\$ROOT and XEPDB1 first

Before running any commands, it helps to understand the two "containers" you'll be switching between. Oracle Database XE uses what's called a multitenant architecture: one container database (CDB) acts as the outer shell, and inside it sits at least one pluggable database (PDB) — **a self-contained database** that holds your actual data.

A useful analogy: think of the CDB as an apartment building, and a PDB as one specific apartment inside it.

- ✓ **CDB\$ROOT** is the building itself — the shared infrastructure. It manages building-wide things like the front door locks (the SYS and SYSTEM administrator accounts) and utilities. You don't live in the lobby; you only pass through it to manage the building or to get to your apartment.
- ✓ **XEPDB1** is your actual apartment — the pluggable database where real work happens. Oracle Database XE creates this one for you automatically during installation. This is where the lecturer account, the students/courses/enrollments tables, and all your course data actually live.

This is why the commands below switch containers partway through: administrative tasks like setting the SYS/SYSTEM password happen at the CDB\$ROOT (building) level, but creating users and tables for coursework happens inside XEPDB1 (the apartment). If you create the lecturer account or a table while still in CDB\$ROOT, it either fails or ends up in the wrong place — so getting the switch right matters.

Note: You will see XEPDB1 by name throughout this guide because it's the default pluggable database Oracle Database XE creates. If your installation uses a different PDB name, substitute that name wherever XEPDB1 appears.

1. Open your command prompt

Enter: `sqlplus /nolog` — this starts SQL*Plus without logging in yet, since at this point you may not know a working username/password to connect with directly.

2. Enter the following commands, one at a time

Run each command below in order. Each line marked "Command:" is what you type; the italic line underneath explains what it does.

Command:

```
ALTER SESSION SET CONTAINER = CDB$ROOT;
```

***What it does:** Switches your session to the root container (CDB\$ROOT). In a multitenant Oracle database, SYS and SYSTEM passwords are managed at the root level, not inside a pluggable database — so this must run first.*

Command:

```
ALTER USER SYS IDENTIFIED BY AUCA2025 CONTAINER = ALL;  
ALTER USER SYSTEM IDENTIFIED BY AUCA2025 CONTAINER = ALL;
```

***What it does:** Sets the password for the SYS and SYSTEM accounts to AUCA2025. CONTAINER = ALL applies the new password across the root container and every pluggable database under it, so the same password works everywhere afterward.*

Command:

```
SELECT username, account_status FROM dba_users WHERE username IN ('SYSTEM','SYS');
```

***What it does:** Confirms both accounts exist and checks their status. You're looking for `account_status = OPEN` — if it says `LOCKED` instead, see the note below on unlocking it.*

Command:

```
ALTER SESSION SET CONTAINER = XEPDB1;
```

What it does: Switches into the XEPDB1 pluggable database. This is the actual working database where course data lives — the lecturer account, student tables, and everything else in Part 1 and Part 2 are created here, not in the root container.

Command:

```
EXIT;
```

What it does: Closes the SQL*Plus session completely. Because this logs you out, Part 1 starts with logging back in — you can't continue straight from here.

Note: If a user is locked, use this query to unlock it: `ALTER USER SYS ACCOUNT UNLOCK CONTAINER = ALL;`

Note: Skip Part 0 if you already did this when installing SQL Developer and you remember the SYS or SYSTEM password.

Part 1: Lecturer Account Creation

3. Log back in as SYS

Part 0 ended with EXIT, so you're no longer connected.

Open Command Prompt and run:

```
sqlplus sys/AUCA2025@localhost:1521/XEPDB1 as sysdba
```

— this logs in directly to the XEPDB1 pluggable database as SYS (using the password you set in Part 0), ready to create the lecturer account below.

1. Create the lecturer account

```
CREATE USER lecturer IDENTIFIED BY AUCA2025;
```

2. Grant necessary privileges to lecturer

```
GRANT CONNECT, RESOURCE TO lecturer;
GRANT CREATE SESSION TO lecturer;
GRANT CREATE TABLE TO lecturer;
GRANT CREATE VIEW TO lecturer;
GRANT CREATE PROCEDURE TO lecturer;
GRANT CREATE SEQUENCE TO lecturer;
GRANT CREATE TRIGGER TO lecturer;
GRANT CREATE PACKAGE TO lecturer;
GRANT UNLIMITED TABLESPACE TO lecturer;
```

3. Create a connection in SQL Developer

4. Click the green "+" icon

This opens the New Connection dialog.

5. Fill in the details

Connection Name: a name you want. Username: lecturer. Password: your password. Hostname: localhost. Port: 1521. SID: XE.

Part 2: Table Creation

Create the STUDENTS table

```
CREATE TABLE students (
  student_id NUMBER PRIMARY KEY,
```

```

first_name VARCHAR2(50) NOT NULL,
last_name VARCHAR2(50) NOT NULL,
email VARCHAR2(100) UNIQUE,
phone VARCHAR2(20),
date_of_birth DATE,
enrollment_date DATE DEFAULT SYSDATE,
gpa NUMBER(3,2) CHECK (gpa BETWEEN 0 AND 4),
status VARCHAR2(20) DEFAULT 'Active'
);

```

6. Create sequence for auto-generating student IDs

```
CREATE SEQUENCE student_seq START WITH 1001 INCREMENT BY 1;
```

7. Insert 50 sample students

```
INSERT INTO students VALUES (student_seq.NEXTVAL, 'Alice', 'Johnson', 'alice.j@email.com',
'111-0001', DATE '2003-05-15', DATE '2023-09-01', 3.8, 'Active');
```

```
INSERT INTO students VALUES (student_seq.NEXTVAL, 'Bob', 'Smith', 'bob.s@email.com', '111-
0002', DATE '2004-03-22', DATE '2023-09-01', 3.2, 'Active');
```

```
INSERT INTO students VALUES (student_seq.NEXTVAL, 'Carol', 'White', 'carol.w@email.com',
'111-0003', DATE '2003-11-08', DATE '2024-01-15', 3.5, 'Active');
```

```
INSERT INTO students VALUES (student_seq.NEXTVAL, 'David', 'Green', 'david.g@email.com',
'111-0004', DATE '2004-07-30', DATE '2024-01-15', 2.9, 'Active');
```

```
INSERT INTO students VALUES (student_seq.NEXTVAL, 'Emma', 'Davis', 'emma.d@email.com',
'111-0005', DATE '2003-12-12', DATE '2024-09-01', 3.9, 'Active');
```

```
INSERT INTO students VALUES (student_seq.NEXTVAL, 'Frank', 'Miller', 'frank.m@email.com',
'111-0006', DATE '2004-02-18', DATE '2023-09-01', 3.4, 'Active');
```

```
INSERT INTO students VALUES (student_seq.NEXTVAL, 'Grace', 'Wilson', 'grace.w@email.com',
'111-0007', DATE '2003-09-25', DATE '2023-09-01', 3.7, 'Active');
```

```
INSERT INTO students VALUES (student_seq.NEXTVAL, 'Henry', 'Moore', 'henry.m@email.com',
'111-0008', DATE '2004-06-14', DATE '2024-01-15', 3.1, 'Active');
```

```
INSERT INTO students VALUES (student_seq.NEXTVAL, 'Ivy', 'Taylor', 'ivy.t@email.com', '111-
0009', DATE '2003-04-03', DATE '2024-09-01', 3.6, 'Active');
```

```
INSERT INTO students VALUES (student_seq.NEXTVAL, 'Jack', 'Anderson', 'jack.a@email.com',
'111-0010', DATE '2004-10-27', DATE '2023-09-01', 2.8, 'Active');
```

```
INSERT INTO students VALUES (student_seq.NEXTVAL, 'Kate', 'Thomas', 'kate.t@email.com',
'111-0011', DATE '2003-08-19', DATE '2023-09-01', 3.9, 'Active');
```

```
INSERT INTO students VALUES (student_seq.NEXTVAL, 'Leo', 'Jackson', 'leo.j@email.com',
'111-0012', DATE '2004-01-11', DATE '2024-01-15', 3.3, 'Active');
```

```
INSERT INTO students VALUES (student_seq.NEXTVAL, 'Mia', 'Martin', 'mia.m@email.com', '111-
0013', DATE '2003-07-23', DATE '2024-09-01', 3.8, 'Active');
```

```
INSERT INTO students VALUES (student_seq.NEXTVAL, 'Noah', 'Lee', 'noah.l@email.com', '111-
0014', DATE '2004-05-06', DATE '2023-09-01', 3.0, 'Active');
```

```

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Olivia', 'Perez', 'olivia.p@email.com',
'111-0015', DATE '2003-12-29', DATE '2023-09-01', 3.7, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Paul', 'Thompson', 'paul.t@email.com',
'111-0016', DATE '2004-09-15', DATE '2024-01-15', 2.7, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Quinn', 'Garcia', 'quinn.g@email.com',
'111-0017', DATE '2003-06-08', DATE '2024-09-01', 3.5, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Ryan', 'Martinez', 'ryan.m@email.com',
'111-0018', DATE '2004-11-21', DATE '2023-09-01', 3.2, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Sara', 'Robinson', 'sara.r@email.com',
'111-0019', DATE '2003-03-14', DATE '2023-09-01', 3.9, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Tom', 'Clark', 'tom.c@email.com', '111-
0020', DATE '2004-08-07', DATE '2024-01-15', 3.1, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Uma', 'Rodriguez', 'uma.r@email.com',
'111-0021', DATE '2003-10-30', DATE '2024-09-01', 3.6, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Victor', 'Lewis', 'victor.l@email.com',
'111-0022', DATE '2004-04-24', DATE '2023-09-01', 2.9, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Wendy', 'Walker', 'wendy.w@email.com',
'111-0023', DATE '2003-02-17', DATE '2023-09-01', 3.8, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Xavier', 'Hall', 'xavier.h@email.com',
'111-0024', DATE '2004-07-11', DATE '2024-01-15', 3.3, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Yara', 'Allen', 'yara.a@email.com',
'111-0025', DATE '2003-09-04', DATE '2024-09-01', 3.7, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Zack', 'Young', 'zack.y@email.com',
'111-0026', DATE '2004-12-28', DATE '2023-09-01', 3.0, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Amy', 'King', 'amy.k@email.com', '111-
0027', DATE '2003-05-20', DATE '2023-09-01', 3.5, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Ben', 'Wright', 'ben.w@email.com', '111-
0028', DATE '2004-11-13', DATE '2024-01-15', 2.8, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Chloe', 'Lopez', 'chloe.l@email.com',
'111-0029', DATE '2003-08-06', DATE '2024-09-01', 3.9, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Dan', 'Hill', 'dan.h@email.com', '111-
0030', DATE '2004-03-29', DATE '2023-09-01', 3.2, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Ella', 'Scott', 'ella.s@email.com',
'111-0031', DATE '2003-07-22', DATE '2023-09-01', 3.7, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Finn', 'Green', 'finn.g@email.com',
'111-0032', DATE '2004-10-15', DATE '2024-01-15', 3.1, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Gina', 'Adams', 'gina.a@email.com',
'111-0033', DATE '2003-06-08', DATE '2024-09-01', 3.6, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Hugo', 'Baker', 'hugo.b@email.com',
'111-0034', DATE '2004-02-01', DATE '2023-09-01', 2.9, 'Active');

```

```

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Iris', 'Gonzalez', 'iris.g@email.com',
'111-0035', DATE '2003-12-24', DATE '2023-09-01', 3.8, 'Active');
INSERT INTO students VALUES (student_seq.NEXTVAL, 'Jake', 'Nelson', 'jake.n@email.com',
'111-0036', DATE '2004-09-17', DATE '2024-01-15', 3.3, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Kara', 'Carter', 'kara.c@email.com',
'111-0037', DATE '2003-04-10', DATE '2024-09-01', 3.7, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Liam', 'Mitchell', 'liam.m@email.com',
'111-0038', DATE '2004-08-03', DATE '2023-09-01', 3.0, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Maya', 'Perez', 'maya.p@email.com',
'111-0039', DATE '2003-11-26', DATE '2023-09-01', 3.5, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Nick', 'Roberts', 'nick.r@email.com',
'111-0040', DATE '2004-05-19', DATE '2024-01-15', 2.8, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Opal', 'Turner', 'opal.t@email.com',
'111-0041', DATE '2003-10-12', DATE '2024-09-01', 3.9, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Pete', 'Phillips', 'pete.p@email.com',
'111-0042', DATE '2004-06-05', DATE '2023-09-01', 3.2, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Queenie', 'Campbell',
'queenie.c@email.com', '111-0043', DATE '2003-03-28', DATE '2023-09-01', 3.7, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Ron', 'Parker', 'ron.p@email.com', '111-
0044', DATE '2004-07-21', DATE '2024-01-15', 3.1, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Sophia', 'Evans', 'sophia.e@email.com',
'111-0045', DATE '2003-09-14', DATE '2024-09-01', 3.6, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Tyler', 'Edwards', 'tyler.e@email.com',
'111-0046', DATE '2004-12-07', DATE '2023-09-01', 2.9, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Unity', 'Collins', 'unity.c@email.com',
'111-0047', DATE '2003-02-28', DATE '2023-09-01', 3.8, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Vince', 'Stewart', 'vince.s@email.com',
'111-0048', DATE '2004-04-21', DATE '2024-01-15', 3.3, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Willow', 'Sanchez',
'willow.s@email.com', '111-0049', DATE '2003-08-14', DATE '2024-09-01', 3.7, 'Active');

INSERT INTO students VALUES (student_seq.NEXTVAL, 'Xander', 'Morris', 'xander.m@email.com',
'111-0050', DATE '2004-11-07', DATE '2023-09-01', 3.0, 'Active');

COMMIT;

```

Create the COURSES table

```

CREATE TABLE courses (
  course_id NUMBER PRIMARY KEY,
  course_name VARCHAR2(100) NOT NULL,
  course_code VARCHAR2(10) UNIQUE,
  credits NUMBER(1) CHECK (credits BETWEEN 1 AND 5),
  instructor VARCHAR2(50),
  department VARCHAR2(50),
  max_students NUMBER DEFAULT 30,

```

```
    semester VARCHAR2(20)
);
```

8. Create sequence for auto-generating course IDs

```
CREATE SEQUENCE course_seq START WITH 101 INCREMENT BY 1;
```

9. Insert 15 courses across different departments

```
INSERT INTO courses VALUES (course_seq.NEXTVAL, 'Introduction to Programming', 'CS101', 3,
'Dr. Johnson', 'Computer Science', 35, 'Fall 2024');
```

```
INSERT INTO courses VALUES (course_seq.NEXTVAL, 'Database Management Systems', 'CS201', 4,
'Dr. Smith', 'Computer Science', 30, 'Fall 2024');
```

```
INSERT INTO courses VALUES (course_seq.NEXTVAL, 'PL/SQL Programming', 'CS202', 3, 'Dr.
Brown', 'Computer Science', 25, 'Fall 2024');
```

```
INSERT INTO courses VALUES (course_seq.NEXTVAL, 'Data Structures', 'CS301', 4, 'Dr.
Wilson', 'Computer Science', 30, 'Fall 2024');
```

```
INSERT INTO courses VALUES (course_seq.NEXTVAL, 'Web Development', 'CS302', 3, 'Dr. Davis',
'Computer Science', 35, 'Fall 2024');
```

```
INSERT INTO courses VALUES (course_seq.NEXTVAL, 'Algorithms', 'CS401', 4, 'Dr. Miller',
'Computer Science', 25, 'Fall 2024');
```

```
INSERT INTO courses VALUES (course_seq.NEXTVAL, 'Software Engineering', 'CS402', 3, 'Dr.
Garcia', 'Computer Science', 30, 'Fall 2024');
```

```
INSERT INTO courses VALUES (course_seq.NEXTVAL, 'Computer Networks', 'CS403', 3, 'Dr.
Martinez', 'Computer Science', 28, 'Fall 2024');
```

```
INSERT INTO courses VALUES (course_seq.NEXTVAL, 'Operating Systems', 'CS404', 4, 'Dr.
Anderson', 'Computer Science', 30, 'Fall 2024');
```

```
INSERT INTO courses VALUES (course_seq.NEXTVAL, 'Machine Learning', 'CS501', 4, 'Dr.
Thomas', 'Computer Science', 20, 'Fall 2024');
```

```
INSERT INTO courses VALUES (course_seq.NEXTVAL, 'Calculus I', 'MATH101', 4, 'Dr. Taylor',
'Mathematics', 40, 'Fall 2024');
```

```
INSERT INTO courses VALUES (course_seq.NEXTVAL, 'Linear Algebra', 'MATH201', 3, 'Dr.
Moore', 'Mathematics', 35, 'Fall 2024');
```

```
INSERT INTO courses VALUES (course_seq.NEXTVAL, 'Business Communication', 'BUS101', 3, 'Dr.
Jackson', 'Business', 40, 'Fall 2024');
```

```
INSERT INTO courses VALUES (course_seq.NEXTVAL, 'Introduction to Psychology', 'PSY101', 3,
'Dr. White', 'Psychology', 45, 'Fall 2024');
```

```
INSERT INTO courses VALUES (course_seq.NEXTVAL, 'English Composition', 'ENG101', 3, 'Dr.
Martin', 'English', 35, 'Fall 2024');
```

```
COMMIT;
```

Create the ENROLLMENTS table

```
CREATE TABLE enrollments (  
    enrollment_id NUMBER PRIMARY KEY,  
    student_id NUMBER REFERENCES students(student_id),  
    course_id NUMBER REFERENCES courses(course_id),  
    enrollment_date DATE DEFAULT SYSDATE,  
    grade VARCHAR2(2),  
    score NUMBER(5,2),  
    status VARCHAR2(20) DEFAULT 'Active',  
    UNIQUE(student_id, course_id)  
);
```

10. Create sequence for auto-generating enrollment IDs

```
CREATE SEQUENCE enrollment_seq START WITH 1 INCREMENT BY 1;  
  
INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1001, 101, DATE '2024-09-02', 'A',  
92.5, 'Active');  
  
INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1001, 102, DATE '2024-09-02', 'A',  
88.0, 'Active');  
  
INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1001, 103, DATE '2024-09-02', NULL,  
NULL, 'Active');  
  
INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1002, 101, DATE '2024-09-03', 'B',  
82.5, 'Active');  
  
INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1002, 104, DATE '2024-09-03', 'B',  
80.0, 'Active');  
  
INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1003, 102, DATE '2024-09-03', 'A',  
90.5, 'Active');  
  
INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1003, 103, DATE '2024-09-03', NULL,  
NULL, 'Active');  
  
INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1004, 103, DATE '2024-09-04', 'C',  
72.0, 'Active');  
  
INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1004, 105, DATE '2024-09-04', 'B',  
85.5, 'Active');  
  
INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1005, 101, DATE '2024-09-04', 'A',  
95.0, 'Active');  
  
INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1005, 102, DATE '2024-09-04', 'A',  
93.5, 'Active');  
  
INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1006, 104, DATE '2024-09-05', 'B',  
84.0, 'Active');  
  
INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1006, 106, DATE '2024-09-05', 'B',  
81.5, 'Active');
```

```
INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1007, 103, DATE '2024-09-05', 'A',
89.0, 'Active');
INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1007, 107, DATE '2024-09-05', NULL,
NULL, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1008, 101, DATE '2024-09-06', 'B',
78.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1008, 108, DATE '2024-09-06', 'C',
75.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1009, 105, DATE '2024-09-06', 'A',
91.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1009, 109, DATE '2024-09-06', NULL,
NULL, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1010, 102, DATE '2024-09-07', 'C',
70.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1011, 103, DATE '2024-09-07', 'A',
94.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1012, 101, DATE '2024-09-08', 'B',
83.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1013, 102, DATE '2024-09-08', 'A',
92.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1014, 104, DATE '2024-09-09', 'B',
79.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1015, 105, DATE '2024-09-09', 'A',
90.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1016, 106, DATE '2024-09-10', 'C',
71.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1017, 103, DATE '2024-09-10', 'B',
86.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1018, 107, DATE '2024-09-11', 'B',
82.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1019, 108, DATE '2024-09-11', 'A',
93.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1020, 101, DATE '2024-09-12', 'B',
80.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1021, 102, DATE '2024-09-12', 'A',
88.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1022, 103, DATE '2024-09-13', 'B',
81.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1023, 104, DATE '2024-09-13', 'A',
91.0, 'Active');
```

```
INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1024, 105, DATE '2024-09-14', 'B',
83.0, 'Active');
INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1025, 106, DATE '2024-09-14', 'A',
89.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1026, 107, DATE '2024-09-15', 'C',
74.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1027, 108, DATE '2024-09-15', 'B',
85.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1028, 109, DATE '2024-09-16', 'C',
73.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1029, 110, DATE '2024-09-16', 'A',
94.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1030, 101, DATE '2024-09-17', 'B',
82.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1031, 102, DATE '2024-09-17', 'A',
90.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1032, 103, DATE '2024-09-18', 'B',
84.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1033, 104, DATE '2024-09-18', 'A',
87.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1034, 105, DATE '2024-09-19', 'C',
76.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1035, 106, DATE '2024-09-19', 'A',
92.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1036, 107, DATE '2024-09-20', 'B',
85.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1037, 108, DATE '2024-09-20', 'A',
88.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1038, 109, DATE '2024-09-21', 'B',
80.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1039, 110, DATE '2024-09-21', 'A',
93.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1040, 101, DATE '2024-09-22', 'C',
72.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1041, 111, DATE '2024-09-22', 'A',
95.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1042, 112, DATE '2024-09-23', 'B',
81.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1043, 113, DATE '2024-09-23', 'A',
89.0, 'Active');
```

```
INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1044, 114, DATE '2024-09-24', 'B',
83.5, 'Active');
INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1045, 115, DATE '2024-09-24', 'A',
91.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1046, 101, DATE '2024-09-25', 'C',
75.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1047, 102, DATE '2024-09-25', 'B',
86.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1048, 103, DATE '2024-09-26', 'A',
92.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1049, 104, DATE '2024-09-26', 'B',
84.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1050, 105, DATE '2024-09-27', 'A',
90.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1001, 111, DATE '2024-09-28', 'B',
85.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1002, 112, DATE '2024-09-28', 'A',
88.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1003, 113, DATE '2024-09-29', 'B',
82.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1004, 114, DATE '2024-09-29', 'C',
74.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1005, 115, DATE '2024-09-30', 'A',
94.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1006, 101, DATE '2024-09-30', 'B',
83.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1007, 111, DATE '2024-10-01', 'A',
91.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1008, 112, DATE '2024-10-01', 'B',
79.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1009, 113, DATE '2024-10-02', 'A',
87.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1010, 114, DATE '2024-10-02', 'C',
71.0, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1011, 115, DATE '2024-10-03', 'A',
93.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1012, 102, DATE '2024-10-03', 'B',
86.5, 'Active');

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1013, 103, DATE '2024-10-04', 'A',
89.5, 'Active');
```

```

INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1014, 111, DATE '2024-10-04', 'B',
84.5, 'Active');
INSERT INTO enrollments VALUES (enrollment_seq.NEXTVAL, 1015, 112, DATE '2024-10-05', 'A',
92.0, 'Active');

COMMIT;

```

Create the EMPLOYEES table

```

CREATE TABLE employees (
  employee_id NUMBER PRIMARY KEY,
  first_name VARCHAR2(50) NOT NULL,
  last_name VARCHAR2(50) NOT NULL,
  email VARCHAR2(100),
  phone VARCHAR2(20),
  hire_date DATE,
  job_title VARCHAR2(50),
  salary NUMBER(10,2),
  commission_pct NUMBER(2,2),
  department VARCHAR2(50),
  manager_id NUMBER REFERENCES employees(employee_id)
);

```

11. Create sequence for auto-generating employee IDs

```

CREATE SEQUENCE employee_seq START WITH 1 INCREMENT BY 1;

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'John', 'Manager',
'john.m@company.com', '555-1001', DATE '2018-01-15', 'Department Head', 85000, 0.05,
'Computer Science', NULL);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Sarah', 'Developer',
'sarah.d@company.com', '555-1002', DATE '2019-03-20', 'Senior Developer', 75000, NULL,
'Computer Science', 1);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Mike', 'Analyst',
'mike.a@company.com', '555-1003', DATE '2020-06-10', 'Data Analyst', 65000, NULL, 'Computer
Science', 1);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Lisa', 'Designer',
'lisa.d@company.com', '555-1004', DATE '2021-02-01', 'UI/UX Designer', 70000, NULL,
'Computer Science', 1);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Tom', 'Engineer', 'tom.e@company.com',
'555-1005', DATE '2019-09-15', 'Software Engineer', 80000, NULL, 'Computer Science', 1);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Emma', 'Coordinator',
'emma.c@company.com', '555-1006', DATE '2020-11-20', 'Project Coordinator', 60000, NULL,
'Business', NULL);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'David', 'Professor',
'david.p@company.com', '555-1007', DATE '2015-08-01', 'Associate Professor', 95000, NULL,
'Mathematics', NULL);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Anna', 'Researcher',
'anna.r@company.com', '555-1008', DATE '2021-05-10', 'Research Assistant', 55000, NULL,
'Psychology', NULL);

```

```

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Robert', 'Admin',
'robert.a@company.com', '555-1009', DATE '2017-12-05', 'Administrative Officer', 50000,
NULL, 'Business', 6);
INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Julia', 'Technician',
'julia.t@company.com', '555-1010', DATE '2022-03-15', 'Lab Technician', 48000, NULL,
'Engineering', NULL);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Kevin', 'Developer',
'kevin.d@company.com', '555-1011', DATE '2020-07-20', 'Junior Developer', 62000, NULL,
'Computer Science', 2);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Maria', 'Designer',
'maria.d@company.com', '555-1012', DATE '2021-09-10', 'Graphic Designer', 58000, NULL,
'Business', 6);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Chris', 'Analyst',
'chris.a@company.com', '555-1013', DATE '2019-04-25', 'Business Analyst', 68000, NULL,
'Business', 6);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Nina', 'Professor',
'nina.p@company.com', '555-1014', DATE '2016-01-30', 'Assistant Professor', 88000, NULL,
'Mathematics', 7);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Paul', 'Engineer',
'paul.e@company.com', '555-1015', DATE '2020-10-05', 'Mechanical Engineer', 72000, NULL,
'Engineering', NULL);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Rachel', 'Counselor',
'rachel.c@company.com', '555-1016', DATE '2018-06-15', 'Career Counselor', 54000, NULL,
'Psychology', 8);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Steve', 'Coordinator',
'steve.c@company.com', '555-1017', DATE '2022-01-20', 'Events Coordinator', 52000, NULL,
'Business', 6);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Olivia', 'Developer',
'olivia.d@company.com', '555-1018', DATE '2021-08-12', 'Full Stack Developer', 78000, NULL,
'Computer Science', 2);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Brian', 'Researcher',
'brian.r@company.com', '555-1019', DATE '2019-11-08', 'Senior Researcher', 82000, NULL,
'Psychology', 8);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Sophie', 'Admin',
'sophie.a@company.com', '555-1020', DATE '2020-02-14', 'HR Administrator', 56000, NULL,
'Business', 9);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Daniel', 'Technician',
'daniel.t@company.com', '555-1021', DATE '2021-07-22', 'IT Technician', 51000, NULL,
'Computer Science', 1);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Amy', 'Professor',
'amy.p@company.com', '555-1022', DATE '2014-03-10', 'Full Professor', 102000, NULL,
'Computer Science', NULL);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'George', 'Engineer',
'george.e@company.com', '555-1023', DATE '2020-09-18', 'Electrical Engineer', 76000, NULL,
'Engineering', 15);

```

```

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Helen', 'Coordinator',
'helen.c@company.com', '555-1024', DATE '2022-05-30', 'Marketing Coordinator', 59000, NULL,
'Business', 6);
INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Ian', 'Developer',
'ian.d@company.com', '555-1025', DATE '2021-12-01', 'Backend Developer', 73000, NULL,
'Computer Science', 2);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Jessica', 'Analyst',
'jessica.a@company.com', '555-1026', DATE '2020-04-15', 'Financial Analyst', 67000, NULL,
'Business', 13);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Kyle', 'Researcher',
'kyle.r@company.com', '555-1027', DATE '2019-08-22', 'Research Scientist', 85000, NULL,
'Engineering', 15);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Laura', 'Professor',
'laura.p@company.com', '555-1028', DATE '2017-05-12', 'Associate Professor', 91000, NULL,
'Psychology', NULL);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Mark', 'Admin', 'mark.a@company.com',
'555-1029', DATE '2021-10-25', 'Office Administrator', 49000, NULL, 'Mathematics', 7);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Nancy', 'Designer',
'nancy.d@company.com', '555-1030', DATE '2022-02-08', 'Web Designer', 64000, NULL,
'Computer Science', 4);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Oscar', 'Developer',
'oscar.d@company.com', '555-1031', DATE '2020-12-15', 'Mobile Developer', 74000, NULL,
'Computer Science', 2);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Patricia', 'Engineer',
'patricia.e@company.com', '555-1032', DATE '2019-06-20', 'Civil Engineer', 77000, NULL,
'Engineering', 15);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Quinn', 'Analyst',
'quinn.a@company.com', '555-1033', DATE '2021-03-18', 'Systems Analyst', 66000, NULL,
'Computer Science', 3);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Rita', 'Coordinator',
'rita.c@company.com', '555-1034', DATE '2022-07-12', 'Student Services Coordinator', 53000,
NULL, 'Psychology', 8);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Sam', 'Technician',
'sam.t@company.com', '555-1035', DATE '2020-05-25', 'Network Technician', 52000, NULL,
'Computer Science', 21);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Tina', 'Professor',
'tina.p@company.com', '555-1036', DATE '2015-11-08', 'Full Professor', 98000, NULL,
'Mathematics', NULL);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Umar', 'Developer',
'umar.d@company.com', '555-1037', DATE '2021-04-22', 'DevOps Engineer', 79000, NULL,
'Computer Science', 2);

INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Vera', 'Researcher',
'vera.r@company.com', '555-1038', DATE '2019-09-30', 'Lab Researcher', 60000, NULL,
'Engineering', 27);

```

```
INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Walter', 'Admin',  
'walter.a@company.com', '555-1039', DATE '2020-08-14', 'Facilities Administrator', 47000,  
NULL, 'Business', 9);  
INSERT INTO employees VALUES (employee_seq.NEXTVAL, 'Yasmin', 'Designer',  
'yasmin.d@company.com', '555-1040', DATE '2022-06-05', 'Product Designer', 71000, NULL,  
'Computer Science', 4);
```

```
COMMIT;
```

Create the DEPARTMENTS table

```
CREATE TABLE departments (  
    department_id NUMBER PRIMARY KEY,  
    department_name VARCHAR2(50) NOT NULL,  
    manager_id NUMBER,  
    location VARCHAR2(50),  
    budget NUMBER(12,2)  
);
```

12. Create sequence for auto-generating department IDs

```
CREATE SEQUENCE dept_seq START WITH 10 INCREMENT BY 10;
```

```
INSERT INTO departments VALUES (dept_seq.NEXTVAL, 'Computer Science', 1, 'Building A -  
Floor 3', 750000);
```

```
INSERT INTO departments VALUES (dept_seq.NEXTVAL, 'Mathematics', 7, 'Building B - Floor 2',  
450000);
```

```
INSERT INTO departments VALUES (dept_seq.NEXTVAL, 'Business', 6, 'Building C - Floor 1',  
550000);
```

```
INSERT INTO departments VALUES (dept_seq.NEXTVAL, 'Engineering', 15, 'Building D - Floor  
4', 800000);
```

```
INSERT INTO departments VALUES (dept_seq.NEXTVAL, 'Psychology', 8, 'Building E - Floor 2',  
400000);
```

```
INSERT INTO departments VALUES (dept_seq.NEXTVAL, 'English', NULL, 'Building F - Floor 1',  
300000);
```

```
COMMIT;
```